

우주비행학 (Space Flight Dynamics)

2018 년 봄학기

Home Work 2

due date: 2018 년 4 월 3 일

- 1) Problem 1(classical Euler angles) in Chapter 4
- 2) In class, we have obtained the following relationships between roll-pitch-yaw Euler angle rates and angular velocity.

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \sin \phi \cos \theta \\ 0 & -\sin \phi & \cos \phi \cos \theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

Solve for roll-pitch-yaw Euler angle rates to get the following equations.

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

- 3) Problem 2 in Chapter 4

- 4) Problem 6 in Chapter 4

* The number of problem is referred by 2nd edition of the test book.

5)

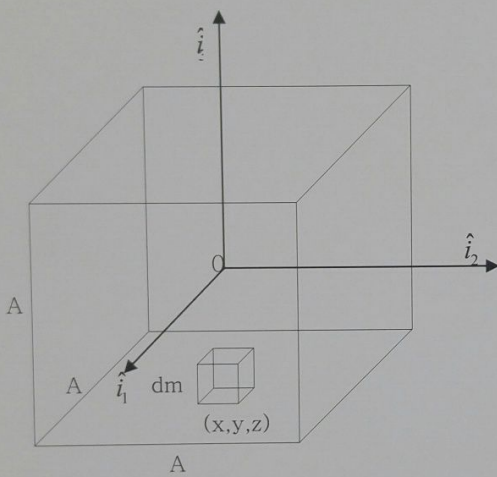
There is a solid cube with side length A and mass density of ρ .
The origin of coordinate is the center of mass.

5.1) Find the moments of inertia.

5.2) Calculate the eigen-values of the moments of inertia

5.3) Find the eigen-vectors corresponding the eigen-values.

5.4) Explain the principal axes.



motion. Kane, Likens, and Levinson is especially noteworthy for its innovative and exhaustive treatment of rotational kinematics, while Hughes' work takes a more conventional approach. Meirovitch has a comprehensive treatment of rotational dynamics using both the classical and advanced techniques of Lagrangian dynamics.

Kane, Thomas R., Likens, Peter W., and Levinson, D. A. *Spacecraft Dynamics*, McGraw-Hill, New York, 1983.

Hughes, Peter C., *Spacecraft Attitude Dynamics*, Wiley, New York, 1986.

Meirovitch, L., *Methods of Analytical Dynamics*, McGraw-Hill, New York, 1970.

4.11 PROBLEMS

1. The angular-velocity component expressions for the Euler angle rates $\dot{\phi}$, $\dot{\theta}$, and $\dot{\psi}$,

$$\omega_1 = \dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi$$

$$\omega_2 = \dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi$$

$$\omega_3 = \dot{\psi} + \dot{\phi} \cos \theta$$

are three linear equations in the three angular rates. Solve for these angular rates, and show that difficulties occur for $\theta \rightarrow 0$.

2. The expression $\mathbf{H} = I\boldsymbol{\omega}^{\text{bi}}$ is still correct when expressed in the unit vectors of an inertial frame, although the moment-of-inertia tensor is no longer constant. Show that the moment-of-inertia matrix obeys

$$\frac{d}{dt} I = \boldsymbol{\omega}^{\text{bi}} \times I = \begin{Bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{Bmatrix} \{I\} \quad (4.98)$$

and that the statement $\mathbf{M} = \dot{\mathbf{H}}$ becomes

$$\mathbf{M} = (\boldsymbol{\omega}^{\text{bi}} \times I)\boldsymbol{\omega}^{\text{bi}} + I\dot{\boldsymbol{\omega}}^{\text{bi}} \quad (4.99)$$

in the inertial frame.

3. Paralleling the development of section 4.3, write the inertial-velocity vector of a mass element dm as

$$\mathbf{v} = \mathbf{v}_{\text{com}} + \boldsymbol{\omega}^{\text{bi}} \times \mathbf{r} \quad (4.100)$$

where $\mathbf{v}_{\text{com}}$ is the translational velocity of the center of mass, and the second term on the right is the rotational velocity with respect to the center of mass. Show that the kinetic energy of the rigid body becomes

$$T = \frac{1}{2} M \mathbf{v}_{\text{com}} \cdot \mathbf{v}_{\text{com}} + \frac{1}{2} \boldsymbol{\omega}^{\text{bi}} \cdot I \boldsymbol{\omega}^{\text{bi}} \quad (4.101)$$

where M is the total mass of the object.

6. After having demolished the axial docking port on his first try, Mario Jones, famed race car driver turned astronaut, is approaching the Skylab space station's bottom port for his second docking attempt at $\mathbf{v} = 10 \text{ m/s } \mathbf{b}_2$. As shown in Figure 4.13, calculate

- (a) the velocity of the center of mass,
- (b) the angular momentum about the center of mass.

Docking is successful (that is, the wreckage stays together). At the moment of contact, two forces arise within the system which the astronauts find hard to ignore, but which you can ignore after you explain *why* they can be ignored. All spacecraft attitude-control devices fail at docking. Calculate

- (c) the moment-of-inertia tensor of the docked configuration,
- (d) the angular velocity of the docked configuration.

The spacecraft masses are $m_a = 15,000 \text{ kg}$ and $m_{sl} = 50,000 \text{ kg}$ for the Apollo and Skylab, and their moment-of-inertia tensors are

$$I_a = \begin{Bmatrix} 20 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 20 \end{Bmatrix} \times 10^3 \text{ kg m}^2 \quad (4.107)$$

$$I_{sl} = \begin{Bmatrix} 90 & 0 & 0 \\ 0 & 50 & 0 \\ 0 & 0 & 50 \end{Bmatrix} \times 10^3 \text{ kg m}^2 \quad (4.108)$$

in the $\mathbf{b}$ frame.

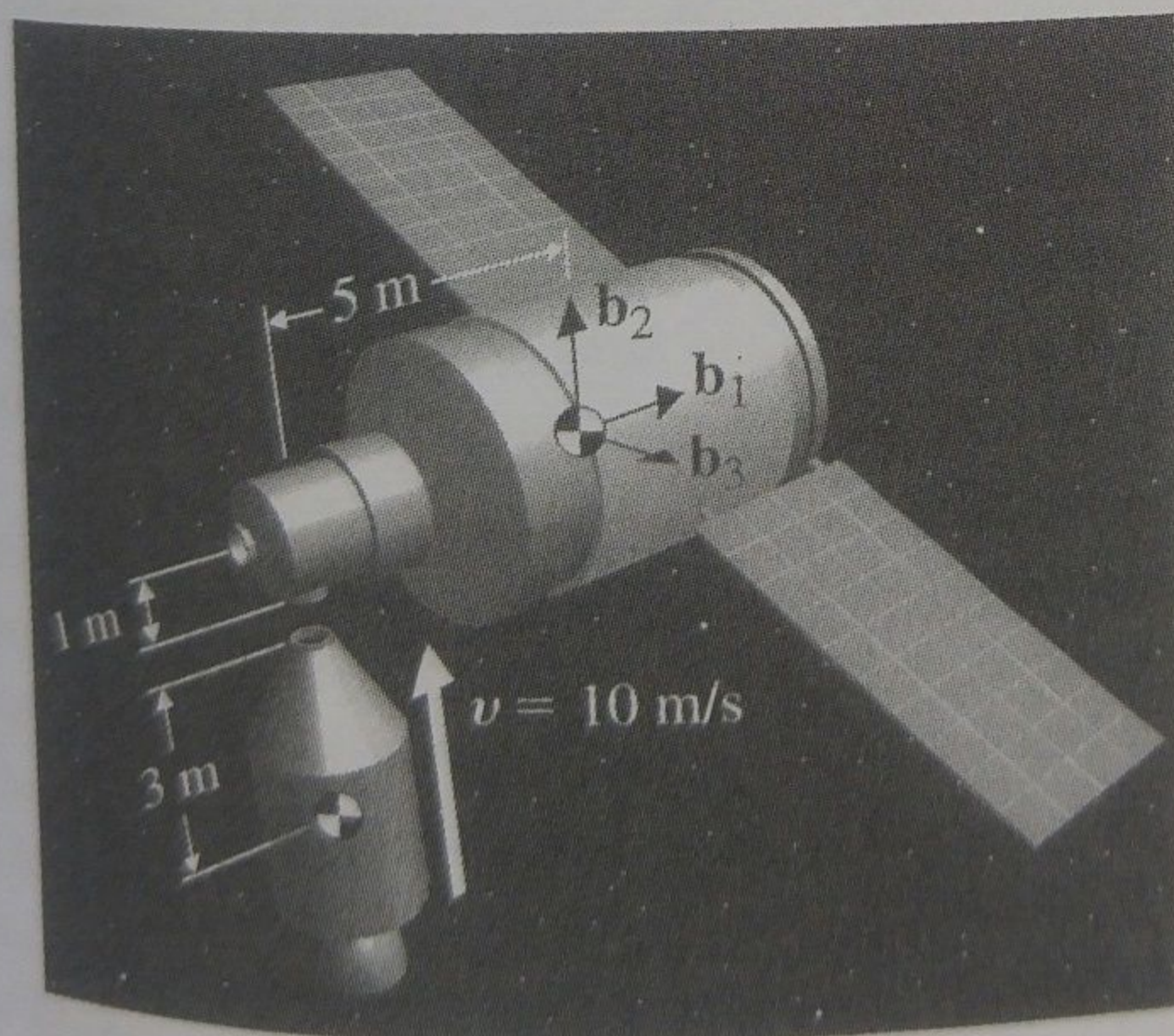


FIGURE 4.13
Nonaxial docking.

7. A thrust-vector control system swivels the rocket engine to maintain correct orientation of the booster. As shown in Figure 4.14, the small engine gimbal angles are θ_1 and θ_2 . Show that the torques are

$$M_1 = lT\theta_1, \quad M_2 = lT\theta_2, \quad M_3 = 0 \quad (4.109)$$

where T is the thrust and l is the distance from the center of mass to the engine. Also, if the control angles are made proportional to the angular rates

$$\theta_1 = K\omega_1, \quad \theta_2 = K\omega_2 \quad (4.110)$$